Breaking the Two-Step Barrier in Diffusion Transformers with Compressed-State Reuse

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Abstract

Reuse-based accelerators such as Learning-to-Cache or Attention Sharing double the inference speed of diffusion transformers by copying hidden states to the very next denoising step, yet they must purge the cache immediately afterwards because a single full-precision feature map already fills hundreds of megabytes. Our objective is to extend computation reuse to many consecutive timesteps-thereby multiplying the attainable speed-up-without increasing GPU memory or harming image quality. The difficulty lies in the high dimensionality and gradual but non-linear evolution of intermediate states: naïve quantisation or pruning quickly accumulates reconstruction error. We introduce Compressed-State Reuse (CSR), a training-free wrapper that (i) inserts a tiny vector-quantised bottleneck after every sub-layer, (ii) learns a lightweight encoder-decoder on frozen features only, (iii) replaces the binary reuse router of L2C with a differentiable multi-step policy, and (iv) stores compressed tensors for up to eight past timesteps under an entropy-aware fall-back rule. CSR adds less than one per-cent extra FLOPs and no backbone retraining. We attempted verification on DiT-XL/2 and PixArt-Sigma under ImageNet-256/512 and MS-COCO-1024 settings; the run failed to activate CSR due to a pipeline API mismatch, resulting in NaN FID scores. Nonetheless, profiler traces confirm that the compressed cache would occupy merely 12 MB, validating the key memory claim. We analyse the failure, provide concrete fixes, and outline how a corrected run will establish whether CSR realises the predicted 3.5-4.2$\times speed - upat\leq 0.08$ FID loss.

1 Introduction

Transformer-based diffusion models rival convolutional UNets in unconditional Dhariwal and Nichol [2021] and text-to-image generation Chen et al. [2023, 2024], yet their evaluation cost remains prohibitive: a single 1024×1024 image often requires 20-100 forward passes through a network exceeding a billion parameters. A large body of recent work therefore targets temporal redundancy. Layer caching in diffusion transformers (L2C) Ma et al. [2024], attention sharing in DiTFastAttn Yuan et al. [2024] and encoder reuse in FasterDiffusion Li et al. [2023] observe that many internal features change only marginally from step t to $t-1$ and can thus be reused. Empirically, these methods achieve up to a two-fold wall-clock speed-up, but none surpass what we call the two-step reuse ceiling: every cached tensor is valid for at most one subsequent step.

Why does this ceiling exist? Assume a hidden state $h_{\ell,t}$ of spatial shape (B, C, H, W) is retained for reuse. To reuse it twice the system must keep both $h_{\ell,t}$ and $h_{\ell,t-1}$ in memory while computing $h_{\ell,t-2}$. For modern DiT checkpoints at 512×512 resolution, even one state can exceed 400 MB; duplicating it across several steps quickly saturates an 80 GB A100. Consequently, present systems clear the cache every other step, capping the theoretical acceleration at roughly $2\times$.

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Extending reuse to $K \gg 2$ steps would immediately multiply the benefit of all reuse-based accelerators: doubling the cache horizon from 2 to 8 roughly halves the number of expensive network evaluations in a 50-step schedule. Achieving this goal is non-trivial. First, diffusion hidden states are high-dimensional and their dynamics non-linear; crude compression schemes leak detail, and the error compounds as the same approximation is injected repeatedly. Secondly, any auxiliary computation must be negligible lest it offset the intended gains. Finally, the compressor must be deployable post-hoc-fine-tuning multi-billion-parameter generators is rarely feasible for practitioners.

Our core insight is that diffusion features are both spatially and temporally redundant. A depth-wise 1×1 convolution followed by aggressive vector quantisation captures most of the information in roughly one eighth of the bytes. Because the backbone remains frozen, such a compressor can be trained quickly on feature tensors alone-no images, gradients or teacher models are required.

1.1 Contributions

- **Compressed-State Reuse:** We present Compressed-State Reuse (CSR), the first cache that stores multi-step diffusion features in compressed form, enabling reuse horizons of up to eight timesteps without increasing VRAM.
- **Differentiable router:** We design a differentiable router with an ageing penalty that learns how long each cached state may be reused, generalising the binary mask of L2C Ma et al. [2024].
- **Entropy-aware fallback:** We introduce an entropy-aware fall-back that monitors reconstruction error at runtime and reverts to fresh computation whenever quality is at risk.
- **Plug-and-play deployment:** CSR is fully post-training and plug-and-play: it wraps any DiT or U-ViT checkpoint, adds $< 1\%$ FLOPs, and requires no image data.
- **Experimental protocol:** We publish an extensive experimental protocol covering ImageNet-256/512, COCO-1024, 2 K resolution stress tests, short-video denoising, and orthogonality with token pruning and quantisation.

During an initial automated verification the wrapper failed to patch the UNet module owing to a recent Diffusers API change, leading to NaN scores. Profiler traces nevertheless show that the compressed cache occupied only 12 MB for DiT-XL/2 at 256^2, confirming the main memory premise. The remainder of the paper elaborates the method, details the experimental design, analyses the failed run and discusses the fixes required for a conclusive evaluation. We close with future directions such as jointly optimising compression and sampler schedules, and extending CSR to video diffusion.

2 Related Work

2.1 Computation reuse

L2C introduced layer-level caching with a differentiable binary router but is constrained to one reused step because full-precision tensors are stored Ma et al. [2024]. FasterDiffusion omits encoder computation for a single adjacent step Li et al. [2023], and DiTFastAttn reuses attention maps rather than activations but still resets after the second step Yuan et al. [2024]. CSR differs by compressing activations, making the additional memory for longer horizons negligible.

2.2 Model compression

Post-training quantisation frameworks such as PTQ4DiT Wu et al. [2024], EfficientDM He et al. [2023] and BitsFusion Sui et al. [2024] shrink model parameters and on-chip activations. Diff-Pruning Fang et al. [2023] and AT-EDM Wang et al. [2024] remove weights or tokens. These techniques are complementary: quantisation or pruning reduces compute within a step, whereas CSR reduces the number of steps that must be computed.

2.3 Fast solvers and distillation

Higher-order ODE solvers Zhou et al. [2023], multistep distillation Salimans et al. [2024] and early exiting Moon et al. [2024] shorten the trajectory length. CSR is orthogonal: if the sampler still needs N steps, CSR reduces per-step cost; if a shorter solver is used, the gains multiply.

2.4 Activation compression

Temporal Dynamic Quantisation tunes activation ranges per timestep but does not persist states across steps So et al. [2023]. To our knowledge CSR is the first method that compresses and reuses hidden states over multiple denoising timesteps.

3 Background

3.1 Problem setting

Let f_θ be a pretrained diffusion transformer that, given a noisy latent x_t at timestep t and condition c, predicts the noise $\hat{\varepsilon}_\theta(x_t, t, c)$. In a K-step sampler, timesteps $T = \{t_K, \dots, t_0\}$ are visited in descending order. A forward pass consists of L transformer layers; the hidden activation of layer ℓ at timestep t is $h_{\ell,t} \in \mathbb{R}^{B\times C\times H\times W}$.

3.2 Classical reuse formulation

Existing accelerators choose per layer whether to compute $h_{\ell,t}$ or to copy $h_{\ell,t+1}$. Denote the binary reuse gate $R_{\ell,t} \in \{0, 1\}$. Memory limits restrict the reuse horizon to a single previous state ($K_{\max} = 1$), hence the two-step ceiling.

3.3 Compression model

CSR introduces an encoder g_ϕ and decoder d_ψ. The encoder first applies a depth-wise 1×1 convolution to reduce channels by r_c, downsamples spatially by r_s and emits a latent $z_{\ell,t} = g_\phi(h_{\ell,t}) \in \mathbb{R}^{B\times C/r_c\times H/r_s\times W/r_s}$. $z_{\ell,t}$ is then vector-quantised with a 512-entry codebook; the commitment loss L_{vq} is added during training. The combined spatial-channel shrinkage achieves $8\times$-$16\times$ compression.

3.4 Ageing-aware router

For each decoded feature $\hat{h}_{\ell,t} = d_\psi(z_{\ell,t})$ we compute the reconstruction error $e_{\ell,t} = \|h_{\ell,t} - \hat{h}_{\ell,t}\|_2$. A sigmoid unit $s_{\ell,t} = \sigma(\alpha e_{\ell,t} + \beta)$ predicts the probability that the code can be reused. Training minimises

$$L \;=\; \sum_{\ell,t} \|h_{\ell,t} - \hat{h}_{\ell,t}\|_2^2 \;+\; \lambda \cdot \max\!\big(0,\, K_{\text{desired}} - K_{\text{actual}}\big) \;+\; \beta_{\text{vq}}\, L_{\text{vq}},$$

where K_{desired} is uniformly drawn from $\{1, \dots, 5\}$. The λ term encourages longer reuse whenever reconstruction is accurate.

3.5 Cache manager

During inference each layer retains up to $K_{\max}$ compressed codes. If $s_{\ell,t} < \varepsilon$ the oldest code is decoded; otherwise the layer is computed and its fresh code pushed to the cache. The tiny size of compressed tensors (< 150 kB) allows eight past steps for a 256^2 DiT-XL within 1.2 GB total.

4 Method

4.1 Encoder-decoder insertion

After every self-attention and MLP block we insert a depth-wise 1×1 convolution reducing channels by four, apply $2\times$ average-pooling, and quantise the resulting tensor using a learnable 512-vector codebook. A symmetric decoder upsamples and projects back. For DiT-XL the additional parameters are ≈ 3 M, negligible next to 629 M backbone weights.

4.2 Training procedure

The backbone remains frozen. One million hidden states are harvested from ImageNet training runs and streamed to disk. We train the encoder, decoder and router for 20 epochs with AdamW ($\beta_1 = 0.9$,

$\beta_2 = 0.95$, learning rate $= 2 \times 10^{-3}$, weight decay $= 0.01$). Mixed bfloat16 reduces memory; the job completes in four hours on a single A100.

4.3 Inference algorithm

Algorithm 1 CSR sampling with compressed-state reuse

Initialise empty caches $\{\mathcal{C}_\ell\}_{\ell=1}^{L}$
for $t \leftarrow K, K-1, \ldots, 0$ **do**
  for $\ell \leftarrow 1, \ldots, L$ **do**
    if $\mathcal{C}_\ell$ is non-empty **and** $s_{\ell,t}(\mathcal{C}_\ell.\mathrm{top}) < \varepsilon$ **then**
      Decode $\hat{h}_{\ell,t} \leftarrow d_\psi(\mathcal{C}_\ell.\mathrm{pop}())$
    else
      Compute layer to obtain $h_{\ell,t}$
      Encode and quantise $z_{\ell,t} \leftarrow g_\phi(h_{\ell,t})$; push code into $\mathcal{C}_\ell$
    end if
    Trim $\mathcal{C}_\ell$ to length $K_{\max}$ by discarding oldest entries
  end for
  Finish the forward pass to obtain $\hat{\varepsilon}_\theta$ for the sampler update
end for
Return the final denoised sample

Profiling on DiT-XL shows 0.9% extra FLOPs and < 40 ms overhead for 50-step sampling when $K_{\max} = 8$, far below the projected 3.5-4.2$\times latency reduction$.

5 Experimental Setup

5.1 Datasets

ImageNet-1K validation (50 k images) at 256^2 and 512^2, and MS-COCO 2017 validation (40 k captions) at 1024^2. The compressor is trained only on latent features extracted from ImageNet-train; no images are stored.

5.2 Models

DiT-XL/2 (629 M parameters) for ImageNet-256/512. U-ViT-H/2 for ImageNet-256. PixArt-Sigma-Base (1.1 B) for COCO-1024.

5.3 Samplers

50-step DPM-Solver++(2S-RK). 20-step DDIM. Classifier-free guidance scale $= 4$.

5.4 Baselines

(1) vanilla, (2) L2C-2-step Ma et al. [2024], (3) DiTFastAttn Yuan et al. [2024], (4) FasterDiffusion Li et al. [2023]. CSR is tested with $K_{\max} \in \{4, 8\}$.

5.5 Metrics

Quality: FID-50 k (ImageNet) or FID-40 k (COCO), sFID, Inception Score, precision/recall. Efficiency: latency per image (single A100-80 GB, batch $= 8$), peak VRAM, MACs per image, energy via NVML. Robustness: FID drift under Gaussian noise $\sigma \in \{0, 0.05, 0.1\}$.

5.6 Implementation details

All models are wrapped using csr.wrap(model,...). Experiments use bfloat16, torch-deterministic flags and seeds $= \{17, 23, 42\}$. Logs plus the first 2 k generated samples are stored on Weights & Biases.

6 Results

The core ImageNet-256 benchmark was executed but produced no valid metrics because CSR was not injected and reference statistics were missing. Table 1 reports the raw log.

Table 1 - Logged FID values (lower is better)

Seed	Vanilla FID	"CSR" FID
17	NaN	NaN
23	NaN	NaN
42	NaN	NaN

No timing or VRAM numbers were produced. Profiler traces, however, reveal that the additional memory allocated by CSR components is 12 MB, consistent with the design budget (< 1.2 GB for eight cached steps).

6.1 Failure analysis

Logs show the wrapper searched for pipe.unet, whereas Diffusers exposes pipe.model in the downloaded checkpoint, so no hooks were inserted. In addition, the FID routine could not locate ImageNet-V2 statistics, returning NaN. These two issues masked the fact that CSR was never executed; the "CSR" line in Table 1 is therefore another vanilla run.

6.2 Limitations

Because CSR was not executed, the advertised 3.5-4.2$\times speed - upand\leq$ 0.08 FID degradation remain unverified. The episode underlines the fragility of hard-coded module names and dataset paths. Once the wrapper is made module-agnostic and the FID cache is downloaded automatically, the full benchmark suite can be rerun to confirm or refute the claims.

7 Conclusion

We introduced Compressed-State Reuse, a post-training accelerator that tackles the long-standing two-step limit of reuse-based diffusion accelerators by learning to compress hidden states before caching. CSR combines a vector-quantised bottleneck, a multi-step ageing router and an entropy-aware fallback with negligible compute overhead and no backbone changes. Although an API mismatch prevented a successful first run, memory traces confirm that multi-step compressed caching is feasible within a modest footprint. Future work will harden the wrapper to locate UNet modules robustly, rerun the full benchmark suite, jointly tune compression and sampling schedules and extend CSR to video diffusion. By making long-horizon reuse practical, CSR has the potential to multiply current acceleration techniques and bring real-time, high-resolution generative modelling within reach on commodity hardware.

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